

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1009	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Pradeepa S	Reg. No.:	192424318
Date:	6/10/2026	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
 - Describe the major functional modules of the proposed system.
 - Identify the functional and non-functional requirements.
 - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
- 5. Justify the Chosen Methodology/Framework**
- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
 - Justify why the selected methodology/framework is suitable for the proposed system.
 - Explain how it supports successful project development and maintenance.
- 6. Discuss Expected Outcomes and Limitations**
- Describe the expected benefits of the proposed solution.
 - Explain how the solution addresses the identified challenges.
 - Discuss potential risks, implementation challenges, and limitations.
 - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity,	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

	scalability, maintainability, security, etc.).			
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Online Course Learning Management System

1. Understand the Problem Statement

Problem Description

The Online Course Learning Management System (LMS) is a software platform designed to provide online education in an organized and flexible manner. The system allows instructors to create and upload courses, learning materials, videos, assignments, and assessments. Students can register for courses, access learning materials, attend assessments, track their learning progress, and receive certificates after successfully completing a course.

The main purpose of the system is to provide a centralized platform where teaching and learning activities can be managed digitally without depending completely on traditional classroom methods.

Objectives

- To provide an online platform for instructors to create and manage courses.
- To allow students to enroll in available courses.
- To provide learning materials such as videos, documents, and assignments.
- To conduct online assessments and quizzes.
- To track student learning progress.
- To generate certificates for successfully completed courses.
- To provide an easy-to-use and secure learning environment.

Scope

The system can be used by colleges, universities, training institutes, and online education providers. It covers course management, student enrollment, learning materials, assessments, progress tracking, and certificate generation. In future, the system can be extended with live classes, discussion forums, AI-based recommendations, mobile applications, and online payment facilities.

2. Identify Key Stakeholders and Challenges

Stakeholders

Stakeholder	Role	Responsibilities
Students	Course Learners	Enroll, learn, attend assessments
Instructors	Course Providers	Upload courses, materials, assessments
Administrator	System Manager	Manage users, courses, certificates

Institution	Platform Owner	Monitor learning activities
-------------	----------------	-----------------------------

Major Challenges

- Managing a large number of students and courses.
- Providing secure access to learning materials.
- Maintaining accurate student progress records.
- Conducting assessments and calculating scores correctly.
- Preventing unauthorized access to courses and certificates.
- Handling large amounts of video and document data.
- Providing a simple and user-friendly interface.
- Maintaining system availability and performance.

3. Analyze the Current Approach

Existing System

In the traditional learning approach, students generally attend physical classes and receive learning materials directly from instructors. Assignments and assessments may be conducted using paper or separate online tools. Student progress and certificates may also be maintained manually.

Some organizations use different platforms for video classes, assignments, assessments, and certificates. This can make the learning process difficult to manage because information is distributed across multiple systems.

Strengths

- Direct interaction between students and instructors.
- Simple for small groups of students.
- Physical classroom activities can be conducted easily.
- Immediate communication is possible during face-to-face classes.

Limitations

- Students must often follow a fixed location and schedule.
- Learning materials may not be available in one centralized location.
- Manual progress tracking can take more time.
- Assessment management may require additional tools.
- Difficult to efficiently manage a large number of students.
- Students may have limited access to learning resources outside class hours.
- Certificate generation and record management may be time-consuming.

Identified Gaps

The existing approach lacks a centralized system for course management, student enrollment, assessments, progress tracking, and certificate generation. Therefore, an integrated Online Course Learning Management System is required.

4. Propose a Suitable Solution Using Software Engineering Principles

Proposed Solution

The proposed system is a web-based Online Course Learning Management System that connects students, instructors, and administrators through a single platform.

The system provides separate functionalities for different users and maintains course, assessment, progress, and certificate information in a centralized database.

Major Functional Modules

Module	Function
User Management	Registration, login, role management
Course Management	Create and upload courses
Enrollment	Student course registration
Assessment	Quizzes and score calculation
Progress Tracking	Monitor course completion
Certificate	Generate certificates
Learning Material Management	Upload and access videos, documents, and assignments

Functional Requirements

- User registration and login.
- Course creation and management.
- Course enrollment.
- Learning material upload and access.
- Online assessments.
- Automatic score calculation.
- Progress tracking.
- Certificate generation.
- User and course management.

Non-Functional Requirements

- **Security:** User information and course data must be protected.
- **Usability:** The system should have a simple and user-friendly interface.
- **Scalability:** The system should support an increasing number of users and courses.
- **Reliability:** The system should provide accurate progress, assessment, and certificate information.
- **Maintainability:** The software should be easy to modify and update.
- **Performance:** Pages, course materials, and assessments should load within reasonable time.

Software Engineering Principles

Modularity:

The system is divided into independent modules such as user management, course management, assessment, progress tracking, and certificates.

Scalability:

The architecture can be designed to support additional students, instructors, courses, and learning materials.

Maintainability:

Separate modules make it easier to identify errors and introduce new features.

Reliability:

The system maintains accurate enrollment, assessment, and progress records.

Usability:

A simple dashboard and clear navigation help students and instructors use the platform easily.

Security:

Authentication and role-based authorization help prevent unauthorized access.

5. Justify the Chosen Methodology / Framework

Selected Methodology: Agile Scrum

Agile Scrum is selected for developing the Online Course Learning Management System because the system contains multiple modules and requirements may change during development. The project can be divided into small sprints.

Sprint-Based Development

Sprint 1: User registration, login, and role management.

Sprint 2: Course creation and course enrollment.

Sprint 3: Learning materials and student dashboard.

Sprint 4: Assessments and score management.

Sprint 5: Progress tracking and certificate generation.

Sprint 6: Testing, security improvements, and final deployment.

Justification

Agile Scrum is suitable because:

- Development can be completed incrementally.
- Requirements can be modified based on user feedback.
- Each module can be developed and tested separately.
- Problems can be identified early.
- Students and instructors can provide feedback during development.
- New features can be added in future iterations.
- Continuous testing improves software quality.

Therefore, Agile Scrum supports both successful development and long-term maintenance of the LMS.

6. Expected Outcomes and Limitations

Expected Outcomes

The proposed system is expected to:

- Provide centralized online learning.
- Reduce manual course management.
- Allow students to learn from anywhere.
- Simplify course enrollment.
- Automate online assessments.
- Provide accurate progress tracking.
- Reduce manual certificate generation.
- Help instructors monitor student performance.
- Improve accessibility and flexibility of education.

How the Solution Addresses the Challenges

The centralized LMS reduces the need for multiple separate platforms. Course materials, assessments, student progress, and certificates are managed within one system. Automated assessment and progress tracking reduce manual work, while authentication and role-based access improve security.

Risks and Implementation Challenges

- Server or network failures may affect access to online courses.
- Large video files may require considerable storage.
- Security vulnerabilities may expose user information.
- A large number of simultaneous users may affect system performance.
- Users may require training to use the system effectively.

Limitations

- The system depends on internet connectivity for online access.
- High-quality video content requires sufficient storage and bandwidth.
- Advanced features may require additional development and infrastructure.
- Automated assessment may not be suitable for all types of examinations.

Future Enhancements

- AI-based personalized course recommendations.
- AI-powered learning assistants and chatbots.
- Mobile application for Android and iOS.
- Live video classes.
- Discussion forums and peer learning.
- Gamification and achievement badges.
- Online payment and premium courses.
- Advanced analytics dashboard for instructors.
- AI-based performance analysis.
- Integration with external educational platforms.

Conclusion

The proposed Online Course Learning Management System provides an integrated solution for managing digital education. It connects students, instructors, and administrators through a centralized platform and supports course management, enrollment, learning materials, assessments, progress tracking, and certificate generation. By following Agile Scrum and software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security, the system can be developed efficiently and enhanced continuously.